# Advanced Fluid Dynamics

## Introduction – DAY 1

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*Dr. Kumaresh*

# Contents


- Introduction
- CFD fundamentals
- Mathematical operations
- Basic Governing Equations
- Installations
- Exercise – 1

# Introduction – About this course

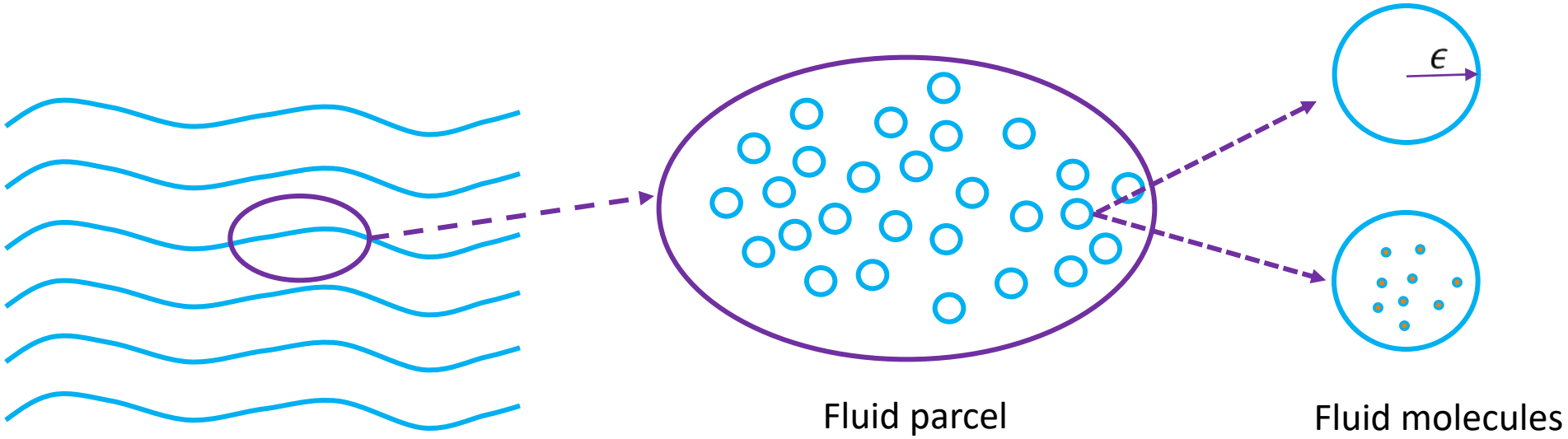
- Course duration per session: 3hrs
- Requirements:
  - Virtual box and installing OS & softwares.
  - Interest to learn CFD using OpenFOAM & Octave
  - Interest to ask questions in discussion forum (GitHub)
  - Work as a team
- Evaluation:
  - Exercises (OpenFOAM and ANSYS Fluent to some extent) (40%)
  - Project Reports (30%)
  - 1 seminar (10%)
  - Attendance (20%)

# References


- Ferziger and Peric; Computational Methods for Fluid Dynamics.
- S. Patankar; Numerical Heat Transfer and Fluid Flow.
- Tannehill et al.; Computational Fluid Mechanics and Heat Transfer.
- Versteeg, Malalasekera; An Introduction to Computational Fluid Dynamics.
- C.J. Greenshields, H.G. Weller; Notes on CFD: General Principles (OpenFOAM)

# CFD fundamentals – Fluid

- A substance whose molecular structure offers no resistance to external forces - Ferziger, Peric



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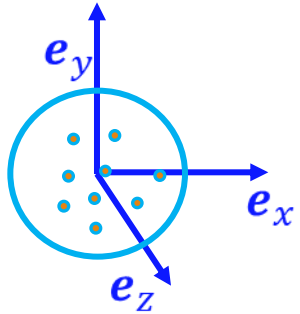
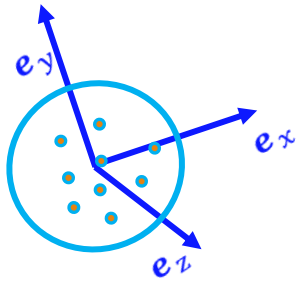
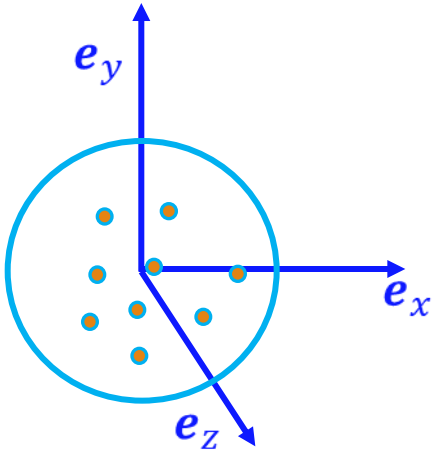


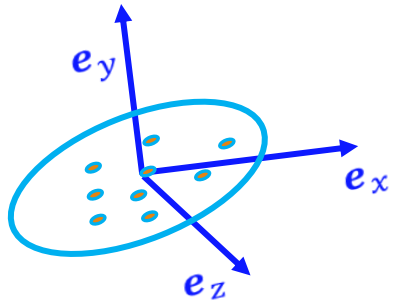

Fluid parcel




Rotation



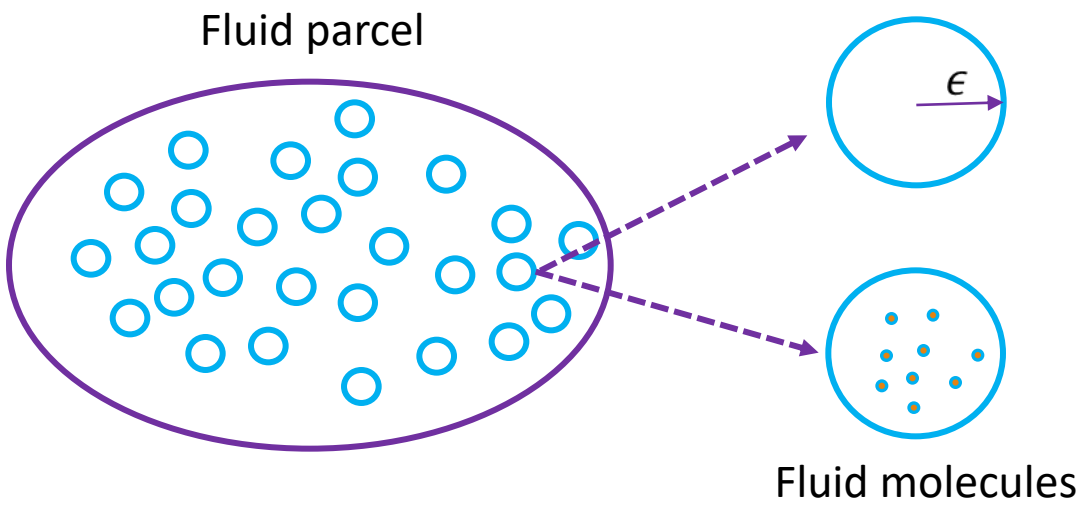

Expansion

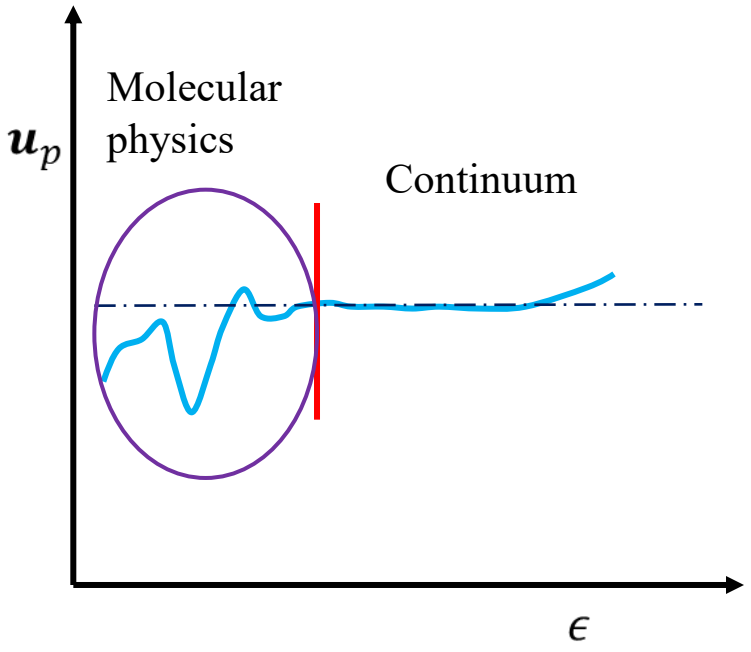



Deformation

# CFD fundamentals – Fluid

- A substance whose molecular structure offers no resistance to external forces - Ferziger, Peric



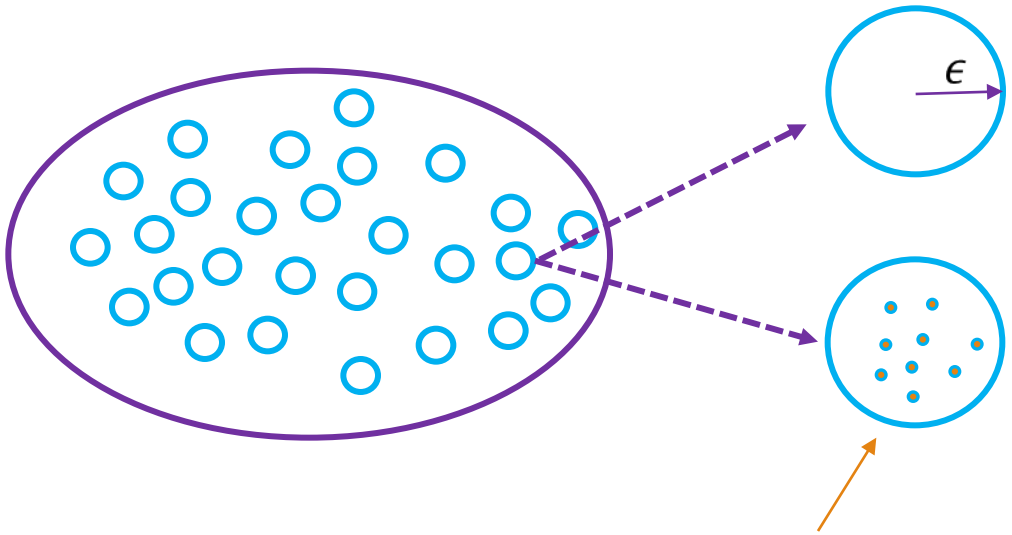

$$\boldsymbol{u}_p = \frac{\sum_{i=1}^{N_{mol}} \boldsymbol{u}_{mol}}{N_{mol}}$$

Fluid velocity: $\boldsymbol{u}(\boldsymbol{x}, t)$

https://en.wikipedia.org/wiki/Knudsen_number#References

# CFD fundamentals – Continuum

Knudsen number: $$\mathrm{Kn} = \frac{\lambda}{L} = \frac{molecular\ mean\ free\ path\ length}{physical\ length}$$

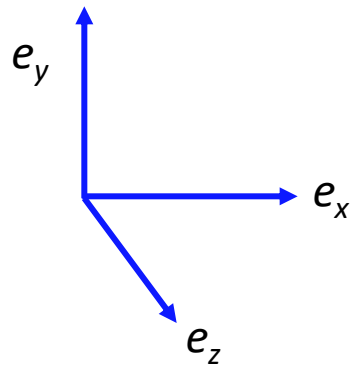



In physics, mean free path is the average distance over which a moving particle

| | |
|---|---|
| Kn < 0.01 | Continuum flow |
| 0.01 < Kn < 0.1 | Slip flow |
| 0.1 < Kn < 10 | Transitional flow |
| Kn > 10 | Free molecular flow |

$$\frac{\partial}{\partial t}(\rho u) + \nabla \bullet (\rho u u) = \nabla \bullet (\mu \nabla u) - \nabla p + S_u$$

# Mathematical operations




Gradient:

$$\nabla \rho = \left( \frac{\partial}{\partial x} \boldsymbol{e}_x + \frac{\partial}{\partial y} \boldsymbol{e}_y + \frac{\partial}{\partial z} \boldsymbol{e}_z \right) \rho = \left( \frac{\partial \rho}{\partial x} \boldsymbol{e}_x + \frac{\partial \rho}{\partial y} \boldsymbol{e}_y + \frac{\partial \rho}{\partial z} \boldsymbol{e}_z \right)$$

[redacted]

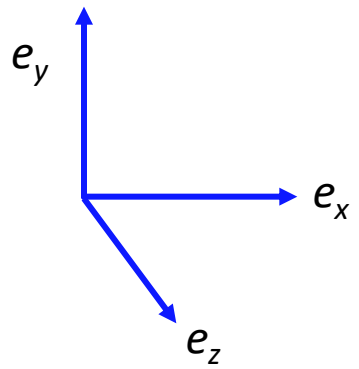
[redacted]

$$\frac{\partial}{\partial t}(\rho u) + \nabla \bullet (\rho u u) = \nabla \bullet (\mu \nabla u) - \nabla p + S_u$$

# Mathematical operations

Gradient:

$$\nabla \rho = \left( \frac{\partial}{\partial x} \boldsymbol{e}_x + \frac{\partial}{\partial y} \boldsymbol{e}_y + \frac{\partial}{\partial z} \boldsymbol{e}_z \right) \rho = \left( \frac{\partial \rho}{\partial x} \boldsymbol{e}_x + \frac{\partial \rho}{\partial y} \boldsymbol{e}_y + \frac{\partial \rho}{\partial z} \boldsymbol{e}_z \right)$$




$$\nabla \boldsymbol{u} = \begin{bmatrix} \partial u/\partial x & \partial v/\partial x & \partial w/\partial x \\ \partial u/\partial y & \partial v/\partial y & \partial w/\partial y \\ \partial u/\partial z & \partial v/\partial z & \partial w/\partial z \end{bmatrix}$$

$$\frac{\partial \rho}{\partial x} = \frac{d\rho}{dx} (in\ 1D)$$

$$\frac{\partial}{\partial t}(\rho u) + \nabla \bullet (\rho u u) = \nabla \bullet (\mu \nabla u) - \nabla p + S_u$$

# Mathematical operations

## Divergence:

- In vector calculus, divergence is a vector operator that operates on a vector field, producing a scalar field giving the **quantity of the vector field's source at each point**.

- More technically, the divergence represents the volume density of the outward flux of a vector field from an infinitesimal volume around a given point.

- As an example, consider air as it is heated or cooled. **The velocity of the air at each point defines a vector field**. While air is heated in a region, it expands in all directions, and thus the velocity field points outward from that region. **The divergence of the velocity field in that region would thus have a positive value**. While the air is cooled and thus contracting, the divergence of the velocity has a negative value.

$$\nabla \cdot \boldsymbol{u} = \left(\frac{\partial}{\partial x}\boldsymbol{e}_x + \frac{\partial}{\partial y}\boldsymbol{e}_y + \frac{\partial}{\partial z}\boldsymbol{e}_z\right)(u\boldsymbol{e}_x + v\boldsymbol{e}_y + w\boldsymbol{e}_z) = \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right)$$

# Governing Equations

- The general equation can be written in the form as:

$$\frac{\partial}{\partial t}(\rho\phi) + \nabla\bullet(\rho u\phi) = \nabla\bullet(\Gamma\nabla\phi) + S$$

$$\frac{\partial}{\partial t}(\rho\phi) + \frac{\partial}{\partial x_j}(\rho u_j\phi) = \frac{\partial}{\partial x_j}\left(\Gamma\frac{\partial\phi}{\partial x_j}\right) + S$$

$$\frac{\partial}{\partial t}\iiint \rho\phi \mathrm{dV} + \iint \rho\phi(u.\mathrm{dA}) = \iint \Gamma\nabla\phi.\ \mathrm{dA} + \iiint S\mathrm{dV}$$

1 2 3 4

1 → Unsteady/Transient term

2 → Advection/Convection term

3 → Diffusion term

4 → Source term

# Governing Equations

$$\frac{\partial}{\partial t}(\rho\phi) + \nabla\bullet(\rho u\phi) = \nabla\bullet(\Gamma\nabla\phi) + S$$

Continuity equation: $\phi = 1, S = 0$ ➔ $\frac{\partial}{\partial t}(\rho) + \nabla\bullet(\rho u) = 0$

Momentum equation: $\phi = u, \Gamma = \mu, S = S_u$ ➔ $\frac{\partial}{\partial t}(\rho u) + \nabla\bullet(\rho \mathrm{u} u) = \nabla\bullet(\mu\ \nabla u) - \nabla p + S_u$

Energy equation: $\phi = h, \Gamma = k / C_p, S = S_h$ ➔ $\frac{\partial}{\partial t}(\rho h) + \nabla\bullet(\rho \mathrm{u} h) = \nabla\bullet\left(\frac{k}{C_p}\ \nabla h\right) + S_h$

Species equation: $\phi = h, \Gamma = \Gamma_l, S = S_m$ ➔ $\frac{\partial}{\partial t}(\rho m_l) + \nabla\bullet(\rho \mathrm{u} m_l) = \nabla\bullet(\ \Gamma_l \nabla m_l) + S_m$

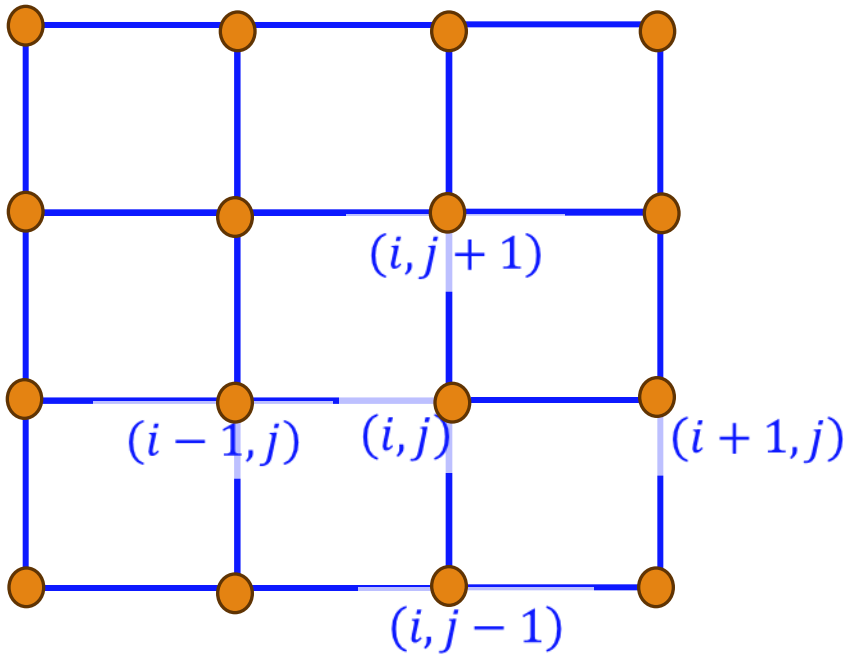
Turbulence equation: $\phi = k(\mathrm{or})\varepsilon, \Gamma = \Gamma_k(\mathrm{or})\Gamma_\varepsilon, S = S_k(\mathrm{or})S_\varepsilon$ ➔ $\frac{\partial}{\partial t}(\rho k) + \nabla\bullet(\rho \mathrm{u} k) = \nabla\bullet(\ \Gamma_k \nabla k) + S_k$

Ideal Gas equation: $p = \rho R T$

# Finite Difference – Finite Volume

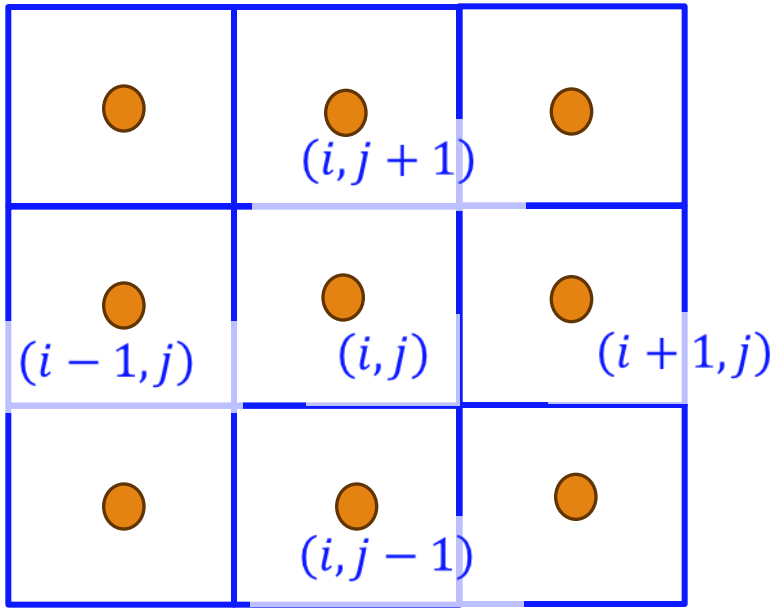
## Differential form

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0$$



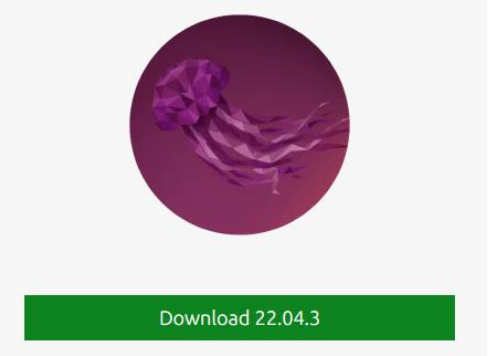

## Integral form

$$\frac{\partial}{\partial t}\int_V \rho dV + \oint_S \rho \boldsymbol{u} \cdot d\boldsymbol{S} = 0$$



# Installations

- List
  - Virtual Box [to create virtual machines]
  - Operating System: Ubuntu 24.04.4 LTS [Install OpenFOAMv2506 & Octave]
  - AnyDesk or RustDesk[For remote access]




OpenFOAM

GNU Octave

- Create a github account:
  - https://github.com/Kumaresh1176/AdvancedFluidDynamics
  - Discussion forum:
    - https://github.com/Kumaresh1176/AdvancedFluidDynamics/discussions

# ANSYS Student version installation

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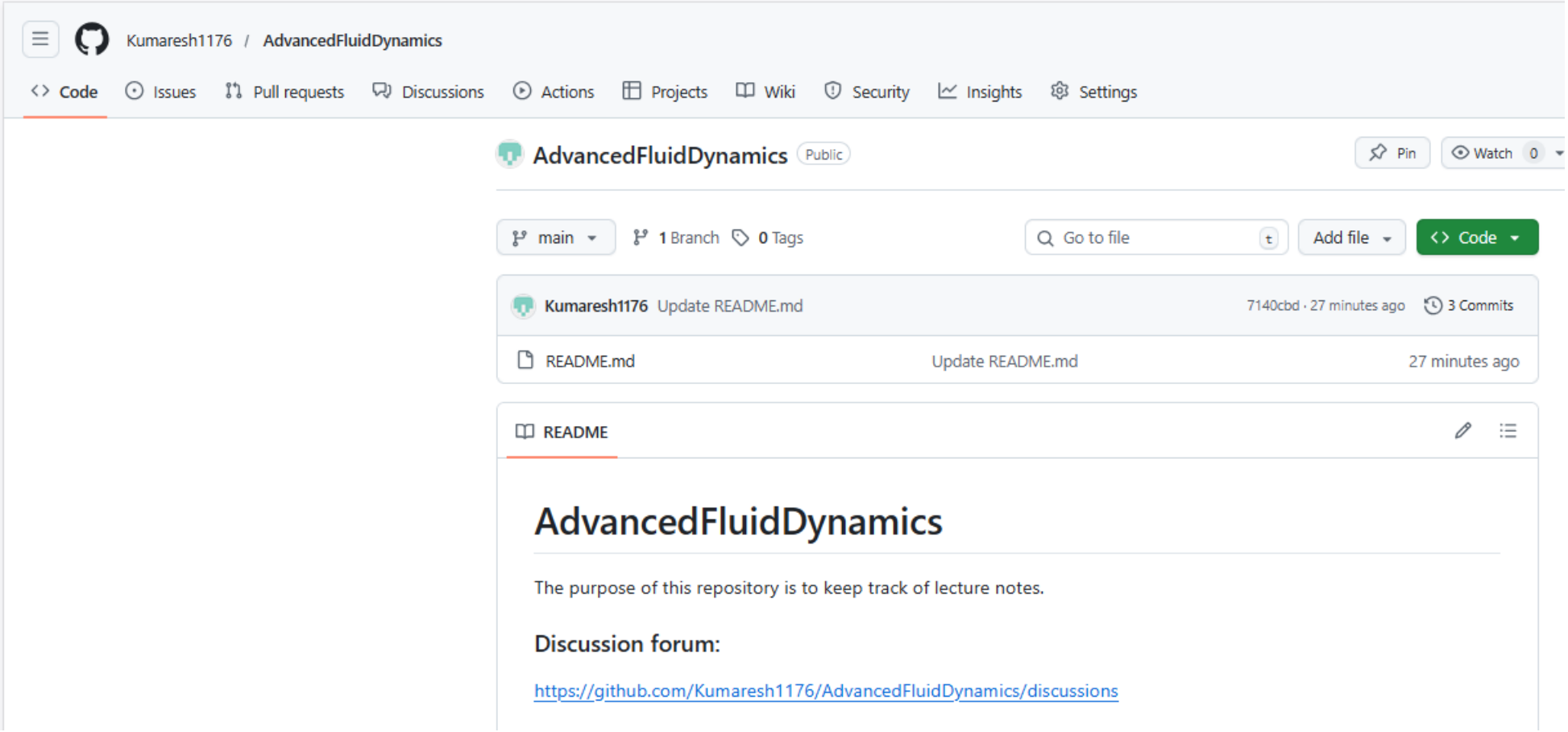
### QUICK LINKS

- Learning Forum
- Innovation Courses
- Learning Resources
- Student Teams

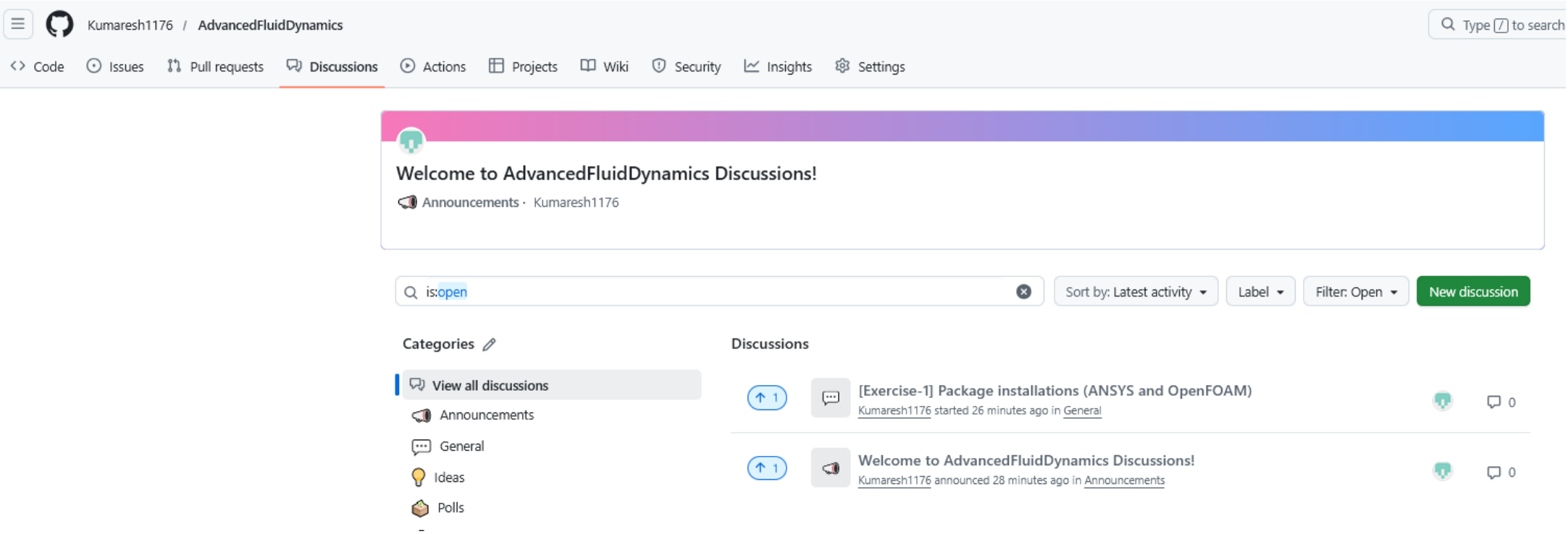
# Presentation and Records

https://github.com/Kumaresh1176/AdvancedFluidDynamics



# Discussions Forum

https://github.com/Kumaresh1176/AdvancedFluidDynamics/discussions



# Introduce yourself here please !

https://github.com/Kumaresh1176/AdvancedFluidDynamics/discussions/1

## Welcome to AdvancedFluidDynamics Discussions! #1

Edit

Kumaresh1176 announced in Announcements

Kumaresh1176 27 minutes ago Maintainer

We're using Discussions as a place to connect with other members of our community. We hope that you:

- Ask questions you're wondering about.
- Share ideas.
- Engage with other community members.
- Welcome others and are open-minded. Remember that this is a community we build together 💪.

To get started, comment below with an introduction of yourself and tell us about what you do with this community.

↑ 1

0 comments

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Paste, drop, or click to add files

Close discussion

Comment

Remember, contributions to this repository should follow our GitHub Community Guidelines.

Category

Announcements

Labels

None yet

1 participant

Notifications

Unsubscribe

You're receiving notifications because you authored the thread.

Lock conversation

Transfer this discussion

Edit pinned discussion

Unpin discussion

Pin discussion to Announcements

Create issue from discussion

Delete discussion

# Exercise – 1

https://github.com/Kumaresh1176/AdvancedFluidDynamics/discussions/2

## [Exercise-1] Package installations (ANSYS and OpenFOAM) #2

Kumaresh1176 started this conversation in General

Kumaresh1176 26 minutes ago Maintainer edited

ANSYS Fluent Package: https://www.ansys.com/products/fluids/ansys-fluent/ansys-fluent-trial

Please install OpenFOAM and GNU Octave based on the documentation:
installation_steps.pdf
install_openfoam_v2506_opt.sh

Follow instructions in installation_steps.pdf

Once after you finish installing, try installing GNU Octave on either Windows or Ubuntu.

Post your queries and make a comment here as part of your Exercise-1.

1

0 comments

Add a comment

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